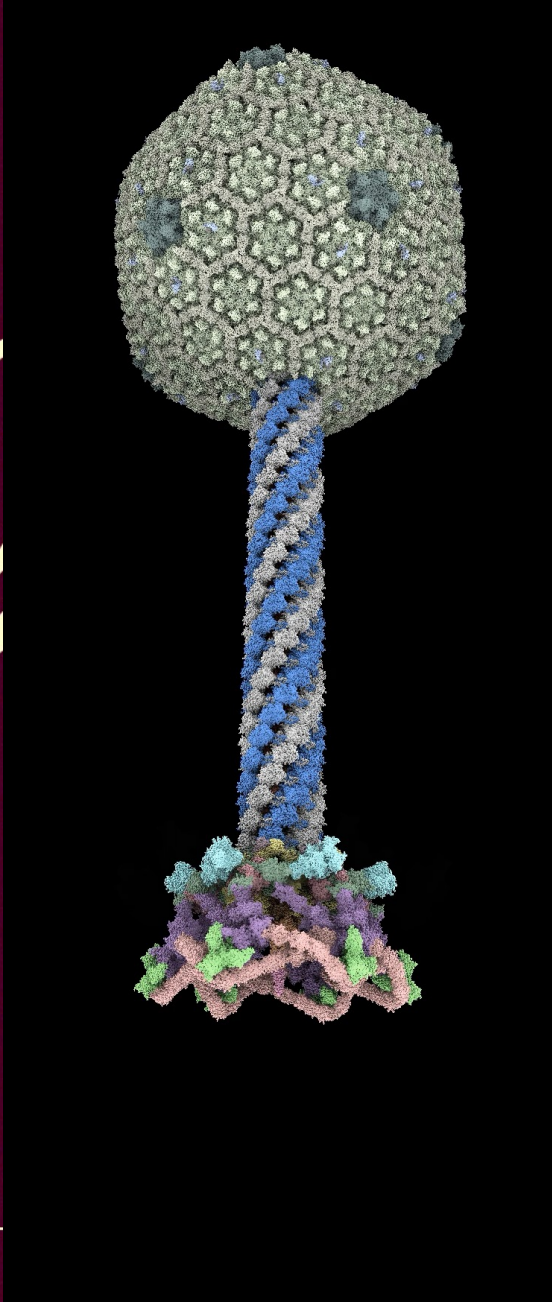


Biochem. 3

nonenzymatic function of proteins



M-CATS @ university of kentucky
9/30 5:00 PM

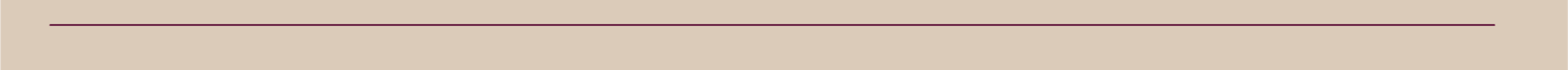
Two questions

Proteins can be enzymes. But what else?

If I had a solution of proteins, how could I isolate one,
test the concentration, identity, and functionality of said protein?

1. Nonenzymatic Protein Function

Proteins can be enzymes. But what else?



5 main types

Structural Proteins

Motor Proteins

Binding Proteins

Biosignaling

Immunoglobulins

Cell Adhesion Molecules

Structural Proteins

Actin

Tubulin

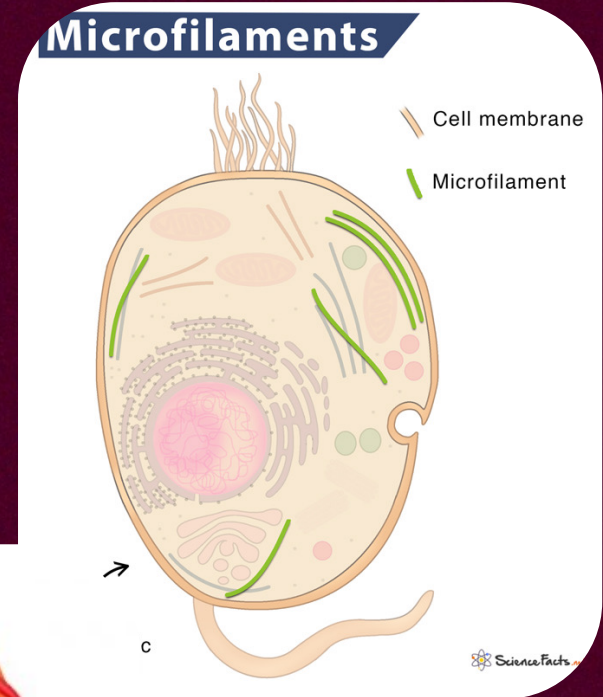
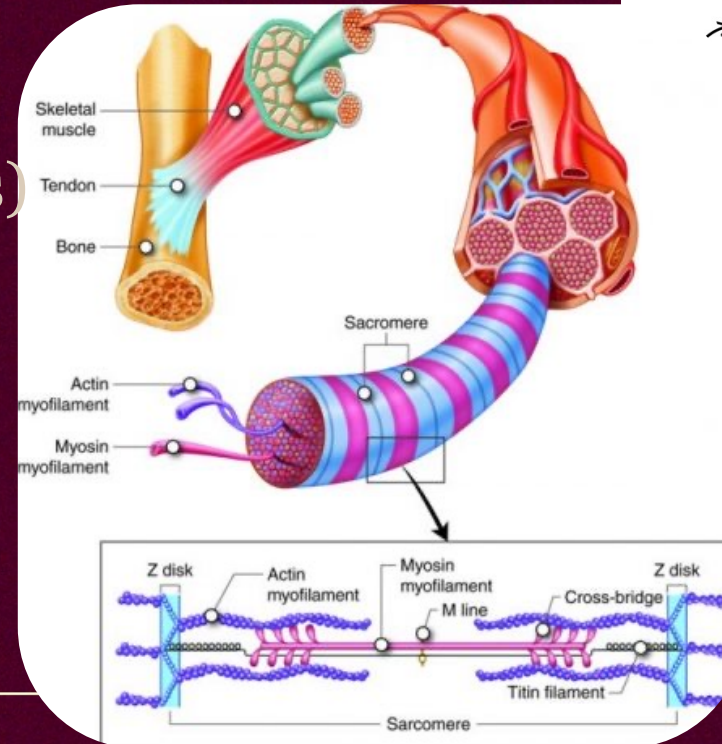
Keratins

Elastins

Collagens

Actin

1. Makes up microfilaments & thin filaments in myofibrils
2. Most abundant protein in eukaryotic cells
3. Have a positive side and negative side (imagine 2 roads that are both one-ways)



Structural Proteins

Actin

Tubulin

Keratins

Elastins

Collagens

Tubulin

1. Makes up microtubules

2. Provides: structure, chromosome

Separation in cell cycle, intracellular transport

With motor proteins (dynein and kinesin)

3. Has polarity:

+ = near periphery

- =near nucleus

Structural Proteins

Actin

Tubulin

Keratins

Elastins

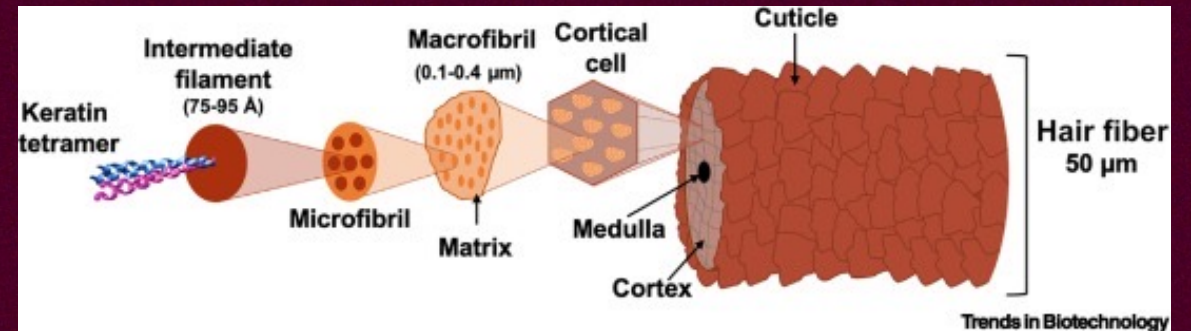
Collagens

Keratins

1. Intermediate filament found in epithelial

2. Makes up hair and nails

3. Contribute mechanical integrity
And function as a regulatory protein



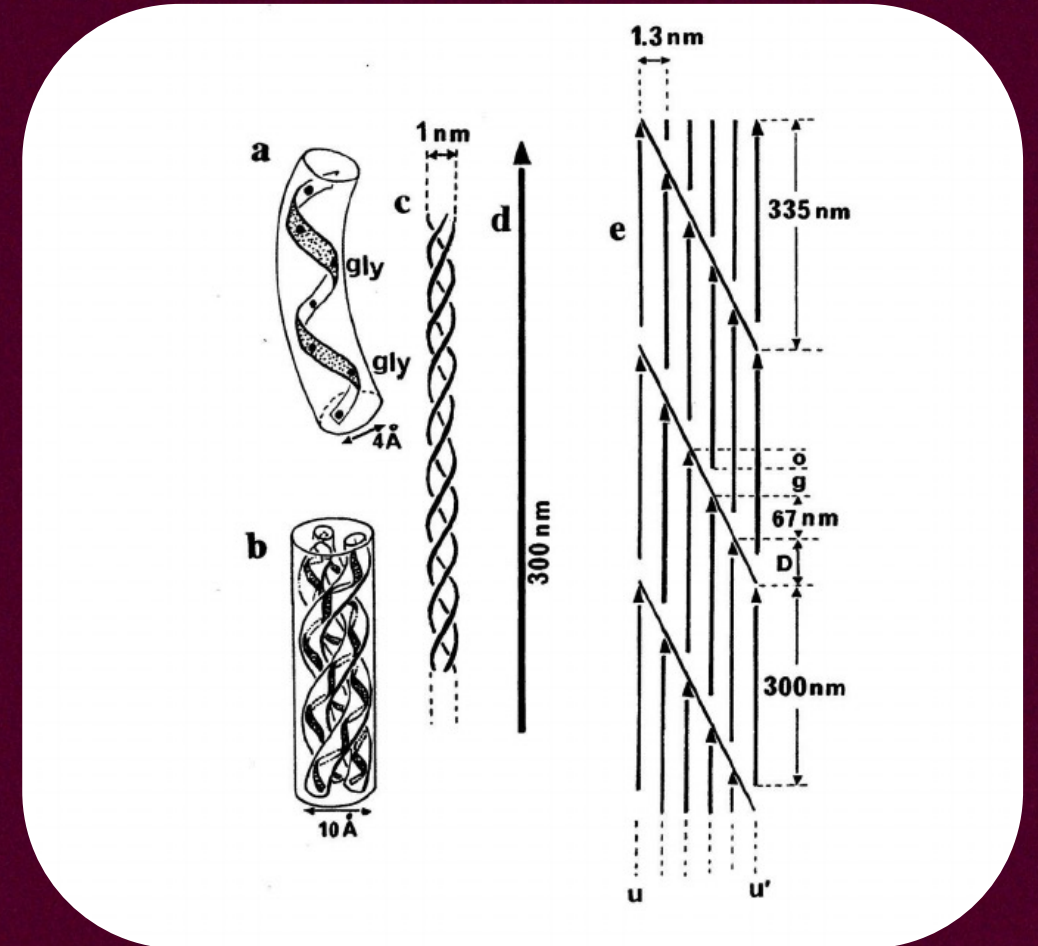
Elastins

1. Makes the extracellular matrix of Connective tissue more elasticky 😊

Collagens

1. Makes up most of Extracellular matrix of connective tissue
2. Provides strength and flexibility
3. 3 left-handed helices woven together to make 1 right-handed helix

Osteogenesis imperfecta



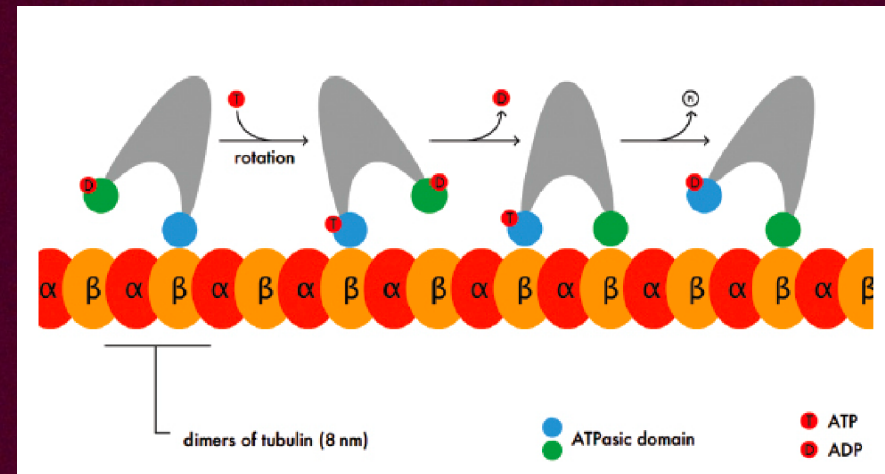
Motor Proteins

Act as ATPases

Myosin

Kinesins

Dyneins



Binding Proteins

Examples: calcium binding, dna binding, hemoglobin

Affinity Curves

Sequestration versus Transportation

Cell Adhesion Molecules

Cadherins

Integrins

Selectins

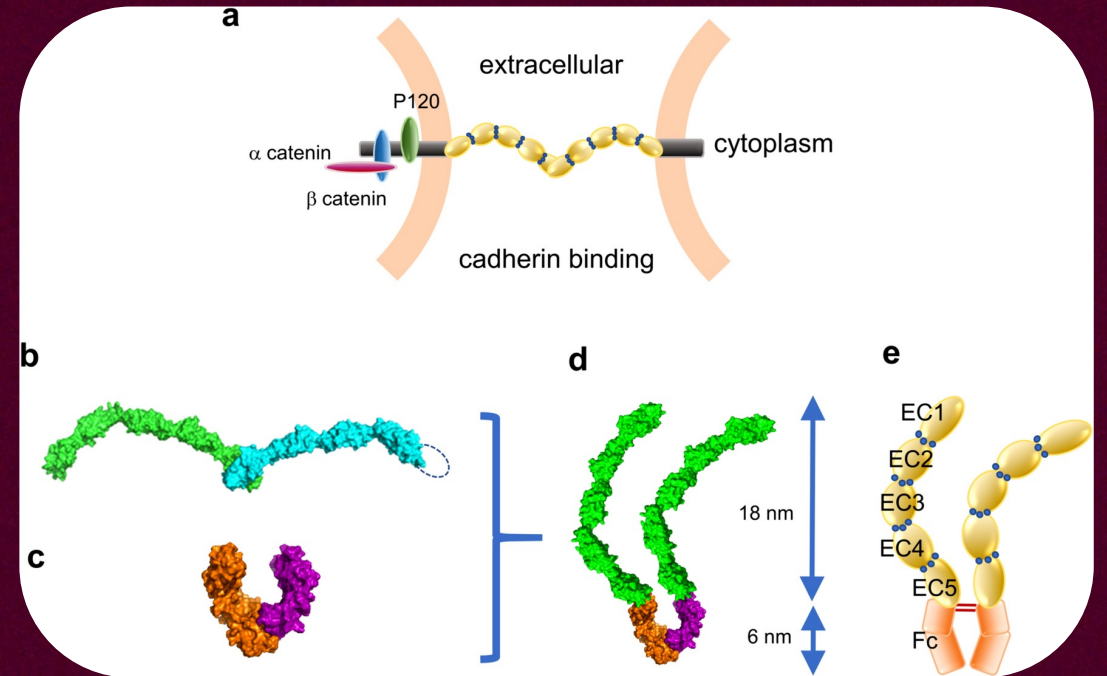
Cadherins

Calcium dependent glycoprotein

Holds similar cell types together

N-Cadherins hold neurons together

E-Cadherins hold epithelial cells together

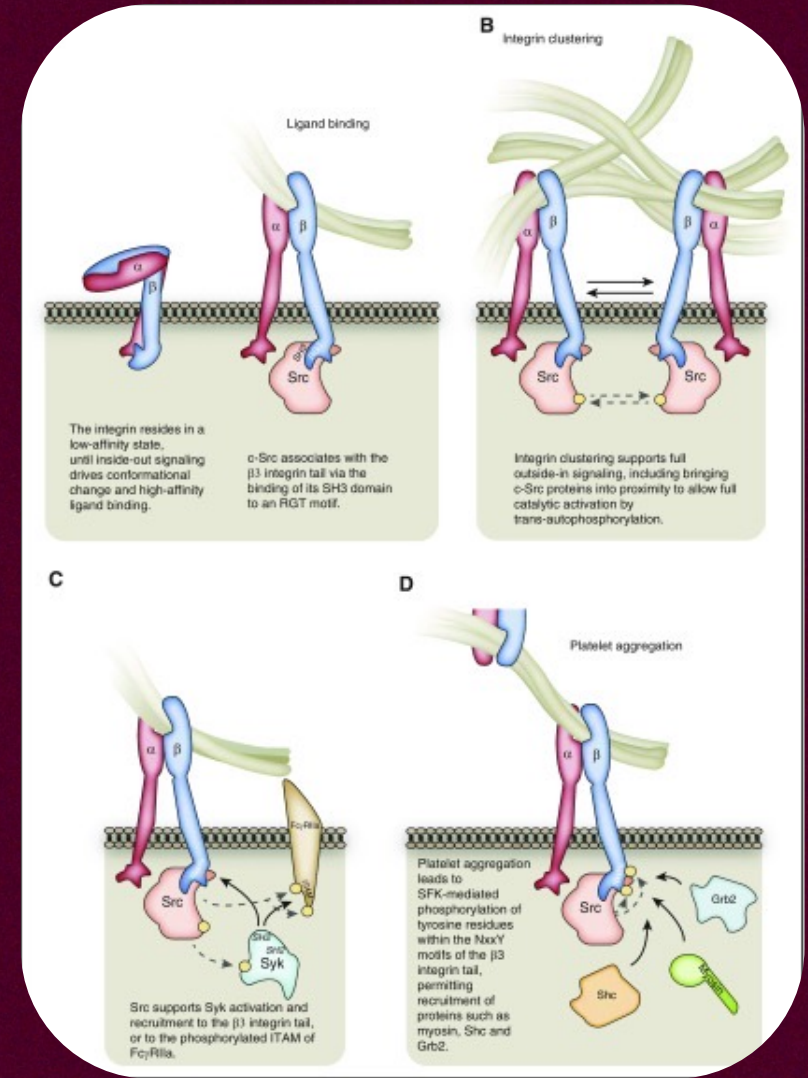


Integrins

Made up of 2 membrane
Spanning domains: alpha and beta

Cellular signaling:
can promote cell division, apoptosis, etc.

Cellular signalling:
can promote cell division, apoptosis, etc.



Selectins

Weakest bond formed

Binds to carbohydrates

Expressed on white blood cells
And endothelial cell lining of blood
vessels



White blood cell go brrrr

Immunoglobulins

Antibody

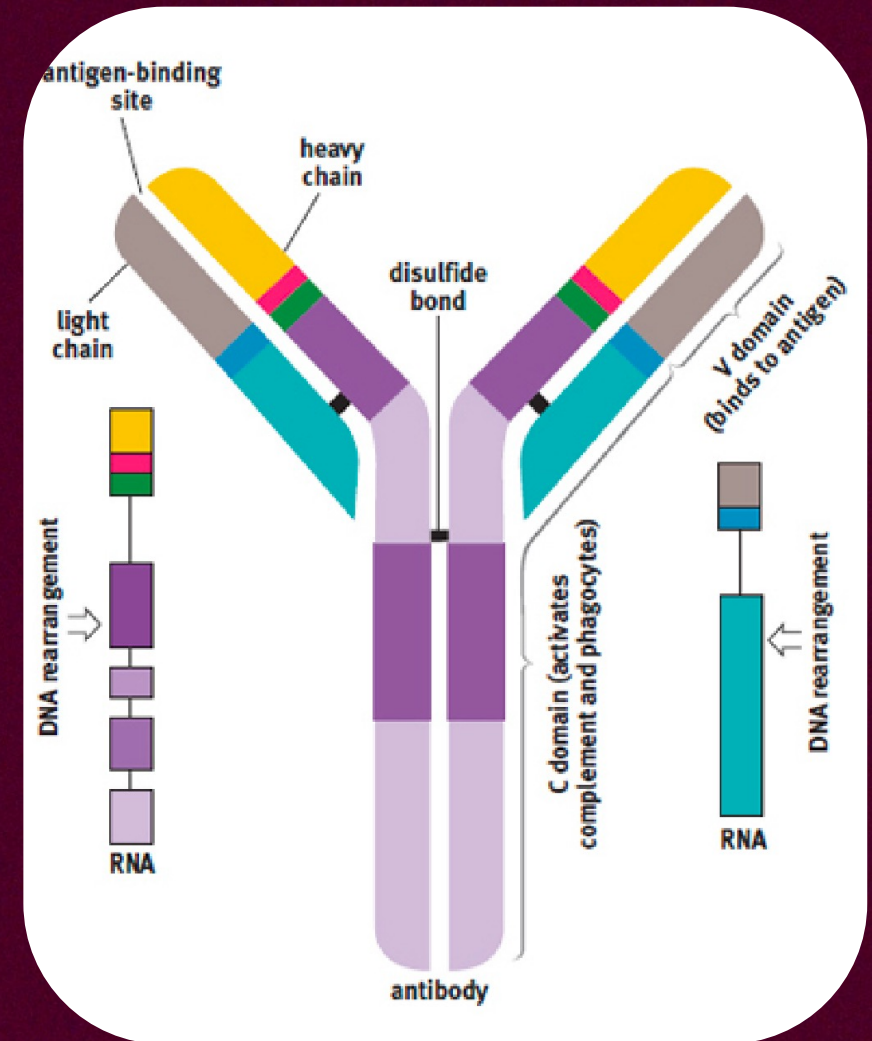
2 Heavy chains & 2 Light chains

Noncovalent interactions and disulfide bonds

Antigen binding site & constant region

3 things could happen when an antibody binds:

1. Neutralize
2. Opsonization
3. Agglutination



Biosignaling

2. *Biosignaling*

This section lowkey big ngl

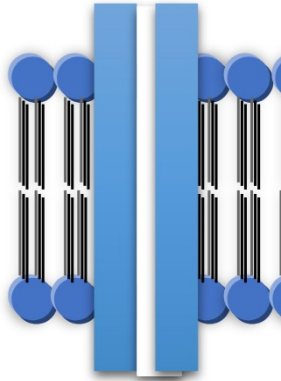
Ion Channels

Ungated Channels

Voltage Gated (NEURONS!! :D)

Ligand Gated (N!! :D)

Nongated
 K^+ Channel



Inside: High K^+

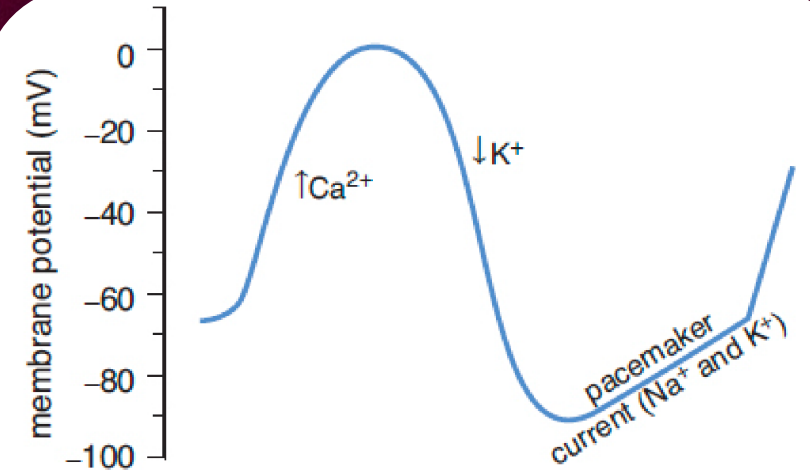
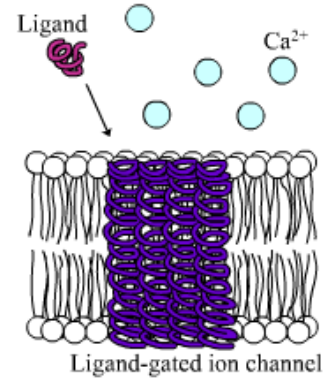


Figure 3.4 Action Potential of the Sinoatrial Node



Ligand-gated ion channel

Enzyme Linked

3 parts:

Ligand binding domain

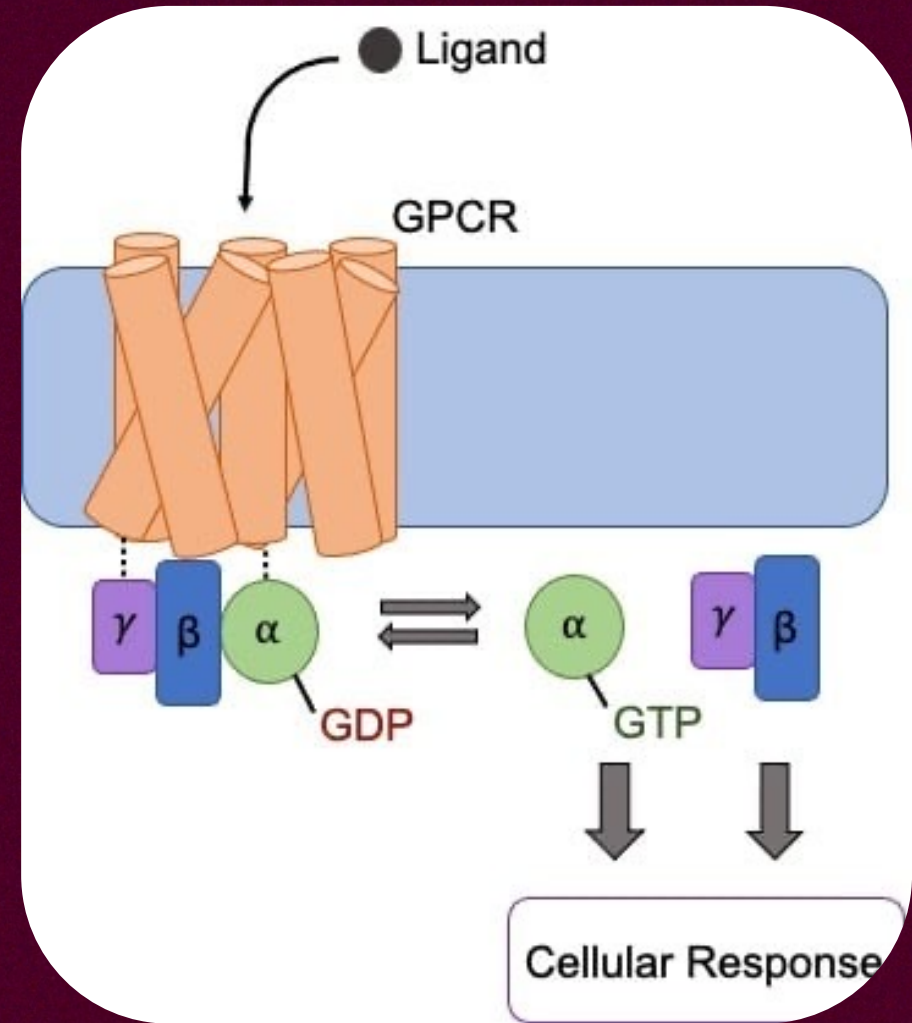
Transmembrane domain

Catalytic Domain → Causes secondary messenger cascade

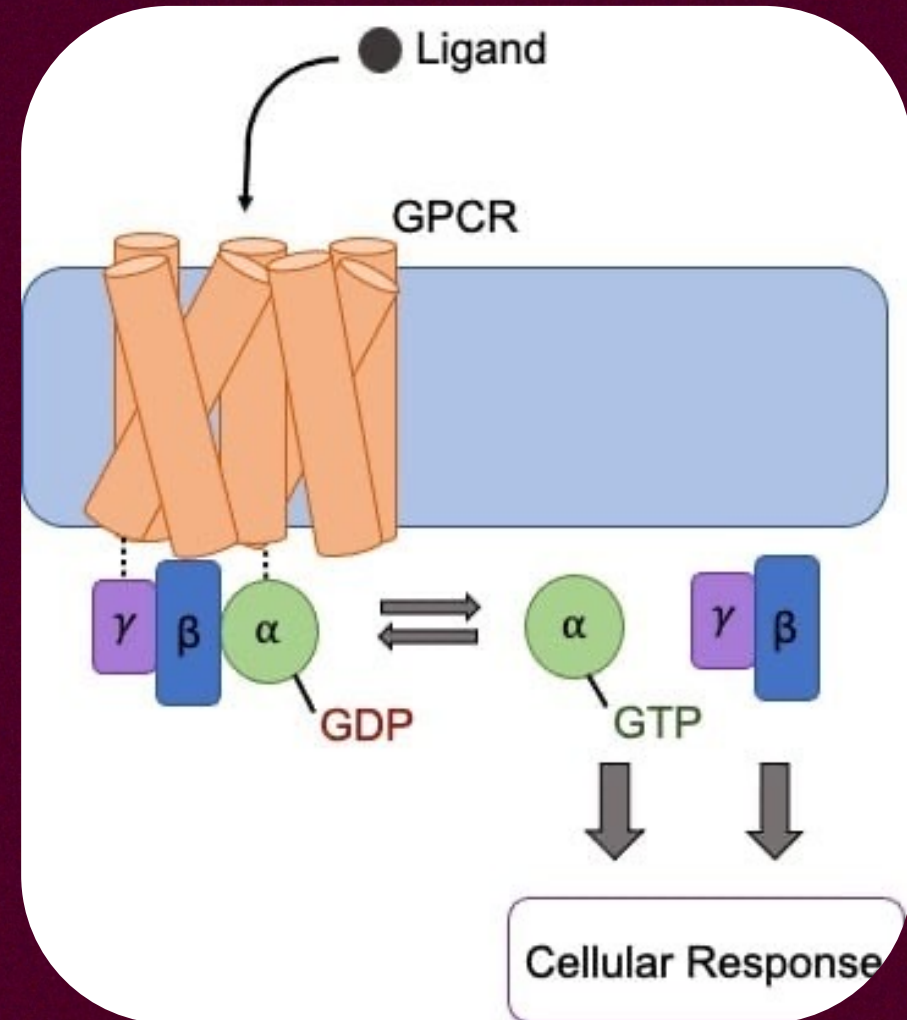
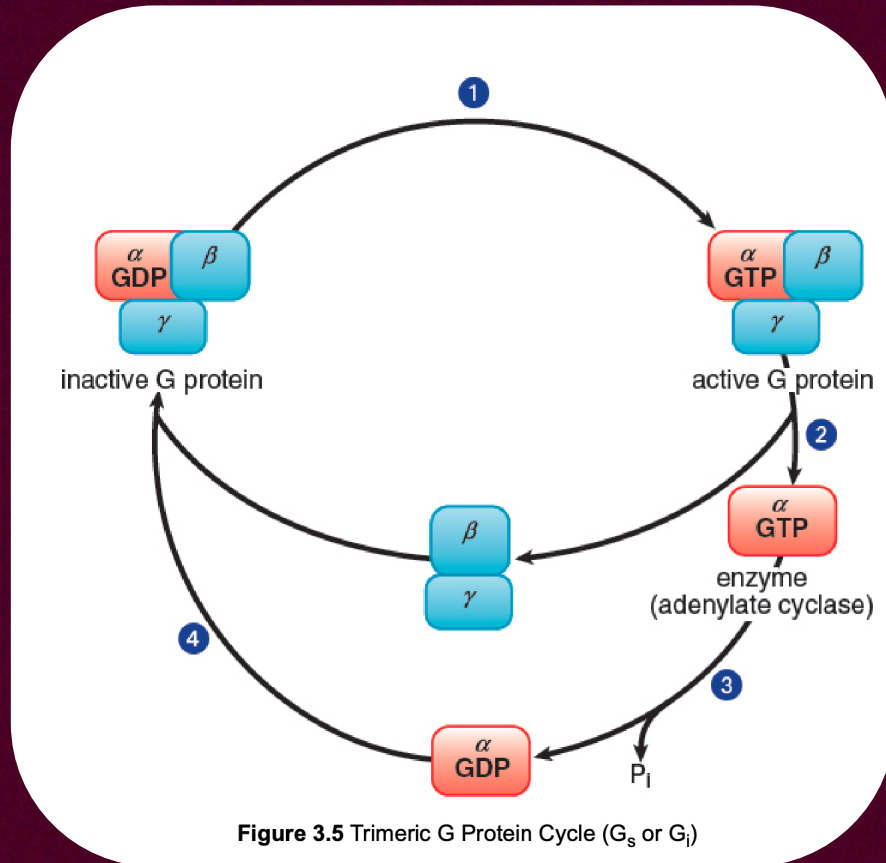
Receptor tyrosine kinases
autophosphorylation

Also serine/threonine kinases,
Receptor tyrosine phosphatases

G protein coupled receptors (GPCRs)



G protein coupled receptors (GPCRs)

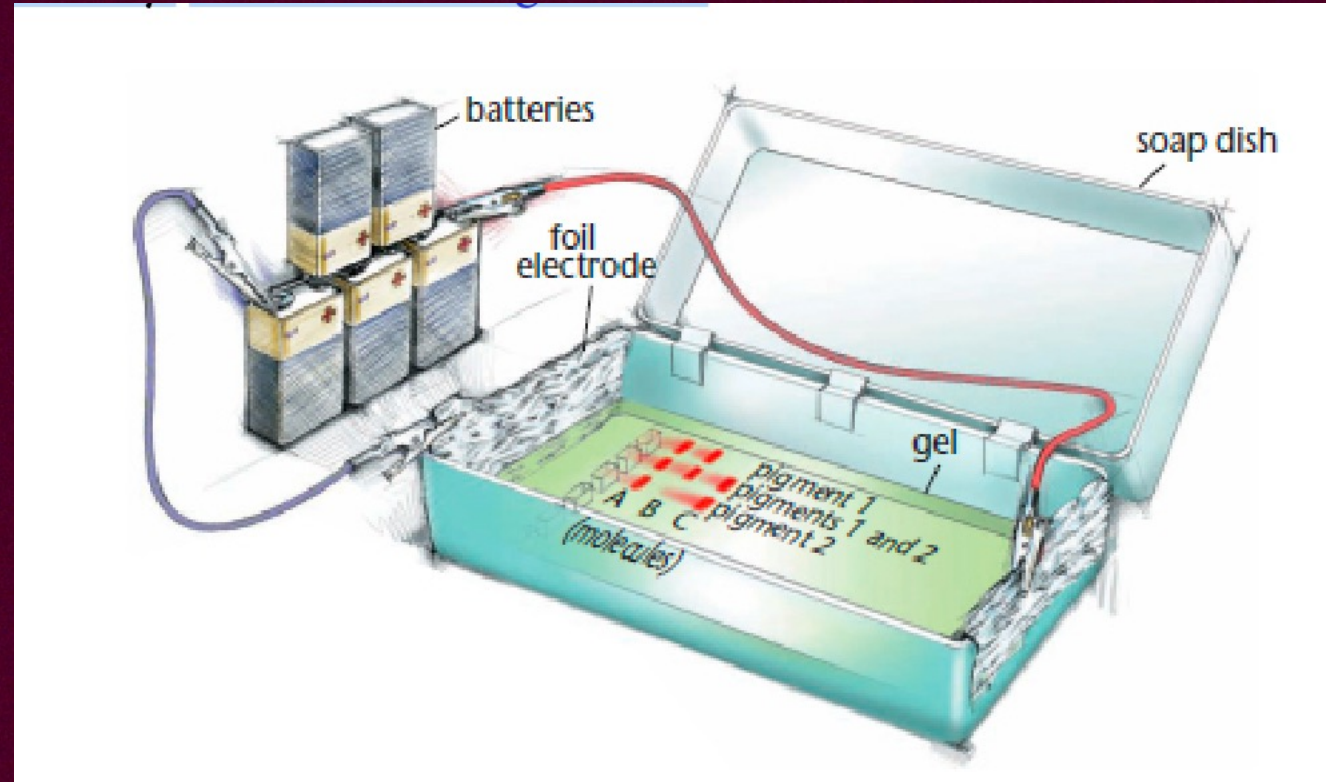


3. Isolating Proteins

We chillin'

Cells → Lysis/homogenization → Centrifugation → Protein mixture → Separation

Electrophoresis



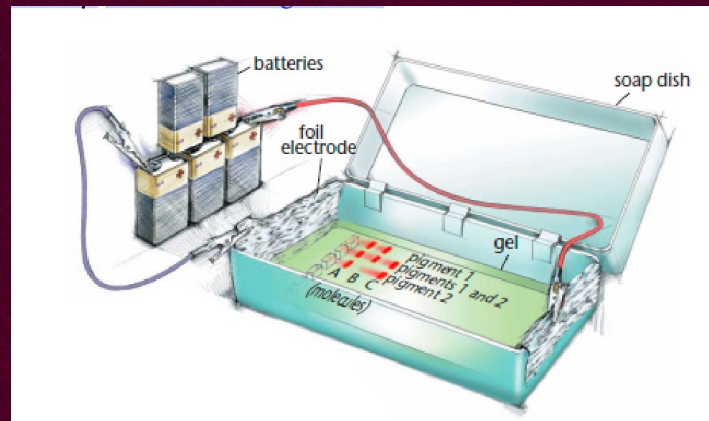
Cells → Lysis/homogenization → Centrifugation → Protein mixture → Separation

Electrophoresis

Native PAGE

SDS PAGE

Isoelectric Focusing

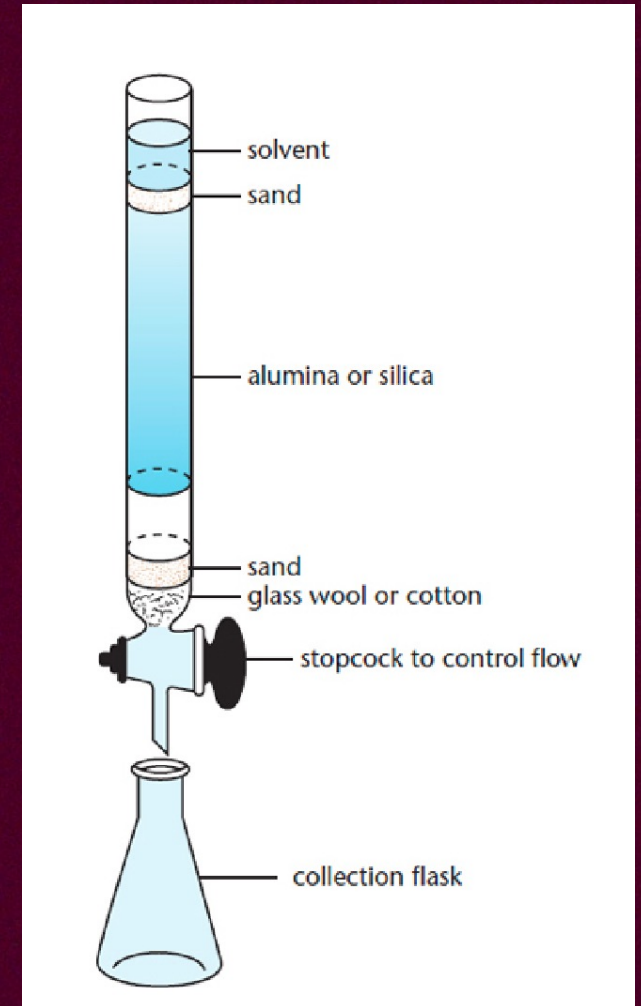


Chromatography

Ion Exchange

Size Exclusion

Affinity



Protein Structure

X-ray

NMR

Protein Structure

X-ray

NMR

Protein Structure

Edman Degradation

Chymotrypsin,
Trypsin,
Cyanogen bromide

Protein Concentration

Lowry Reagent Assay

BCA

Bradford protein assay

